



Milestone One

KVM-Based eBPF Malware Detection Engine

Project Team & Overview

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Faculty Advisor: Dr. Ribeiro

Client: Amy Alvarez, NASA Scientist

Advisor meetings: monthly, TR 2:00–3:15pm (office hours) • Client meetings: monthly, when available

Milestone Overview

- Laying foundation for future work
- Web UI design
- Easy user setup and interaction w/ tool
- Base VMI

Web Server

Two Views

- Dashboard - anything wrong in one line
- Introspection - raw eBPF data for closer inspection

Website updates in real time

Server scans each guest VM on a timer and pushes new findings to browser

Swappable engine

The site is built to talk to a clean interface to allow for easy adjustments as we build the tool

Python Web Server

FastAPI

Web UI

RootView

Home

Dashboard

Introspection

API

live

!

4 findings need attention

RootView found eBPF activity in a guest that does not look legitimate.

GUESTS UNDER INTROSPECTION

web-frontend

RUNNING

Kernel6.8.0-40-generic

Archx86_64

Profileloaded

3 programs / 4 maps

Inspect →

db-primary

RUNNING

Kernel6.5.0-27-generic

Archx86_64

Profileloaded

3 programs / 2 maps

Inspect →

build-runner

RUNNING

Kernel5.15.0-91-generic

Archx86_64

Profilenot loaded

3 programs / 2 maps

Inspect →

archive-store

SHUT OFF

Kernel6.1.0-18-amd64

Archx86_64

Profilenot loaded

- programs / - maps

Inspect →

FINDINGS (4)

MEDIUM

ci_file_trace was loaded by uid 1000

17:44:04

Program 31 (ci_file_trace) is a tracepoint program but was loaded by uid 1000 rather than root. Tracing programs normally require elevated capabilities to load.

What to do: Check what capabilities that account holds and how it obtained them.

► Evidence

LOW

runner_net has no identifiable loader

17:44:04

RootView

Home

Dashboard

Introspection

API

live

EBPF STATE

Guestweb-frontend

Read directly from guest kernel memory at 2026-09-21 17:44:49 UTC. Rows marked HIDDEN exist in the kernel but are not reported by the guest's own tooling.

Programs (3)

ID	NAME	TYPE	ATTACHED TO	LOADER	HELPERS	TAG	VISIBILITY
14	xdp_lb_redirect	xdp	eth0	katran-loader uid 0	bpf_map_lookup_elem bpf_redirect_map bpf_csum_diff	a3f2118c4d90ee71	VISIBLE
15	cgroup_egress	cgroup_skb	/sys/fs/cgroup/system.slice	systemd uid 0	bpf_map_update_elem	77b0cc1e2a5d3f08	VISIBLE
16	exec_audit	tracepoint	sched:sched_process_exec	falco uid 0	bpf_get_current_pid_tgid bpf_perf_event_output	0d41c7ba6659e2a4	VISIBLE

Maps (4)

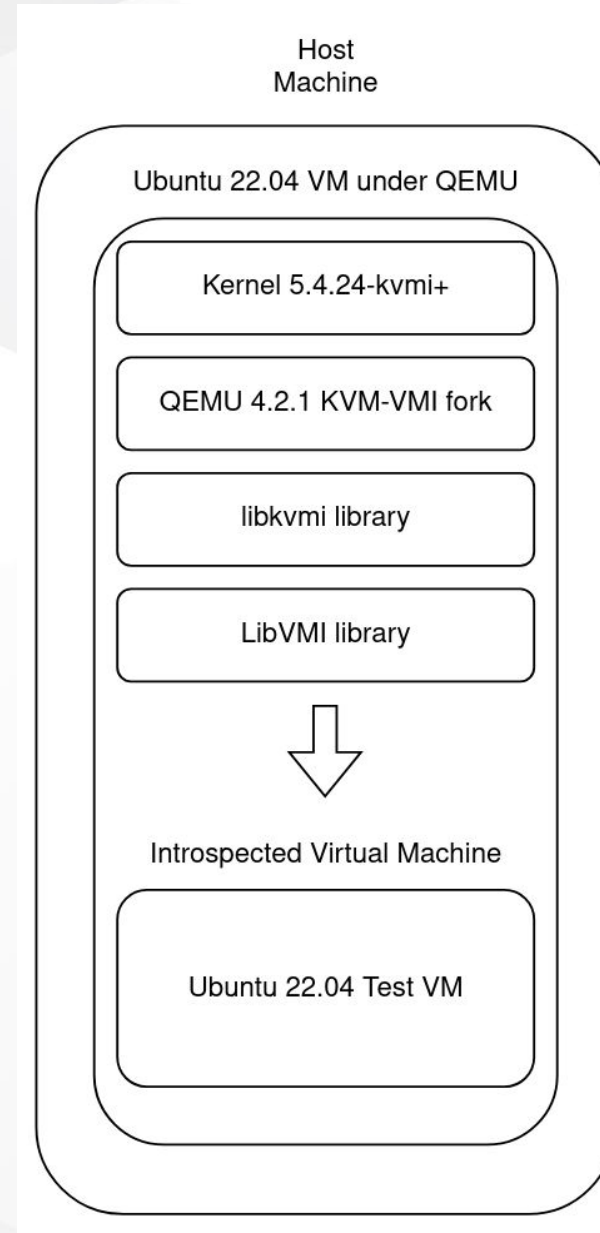
ID	NAME	TYPE	KEY	VALUE	MAX ENTRIES	VISIBILITY
21	lb_backends	hash	8B	16B	4096	VISIBLE
22	xdp_tx_ports	prog_array	4B	4B	64	VISIBLE
23	cgroup_counters	percpu_array	4B	8B	256	VISIBLE
24	exec_events	perf_event_array	4B	4B	8	VISIBLE

backend: demo (FAKE DATA)

scan interval: 2.0s

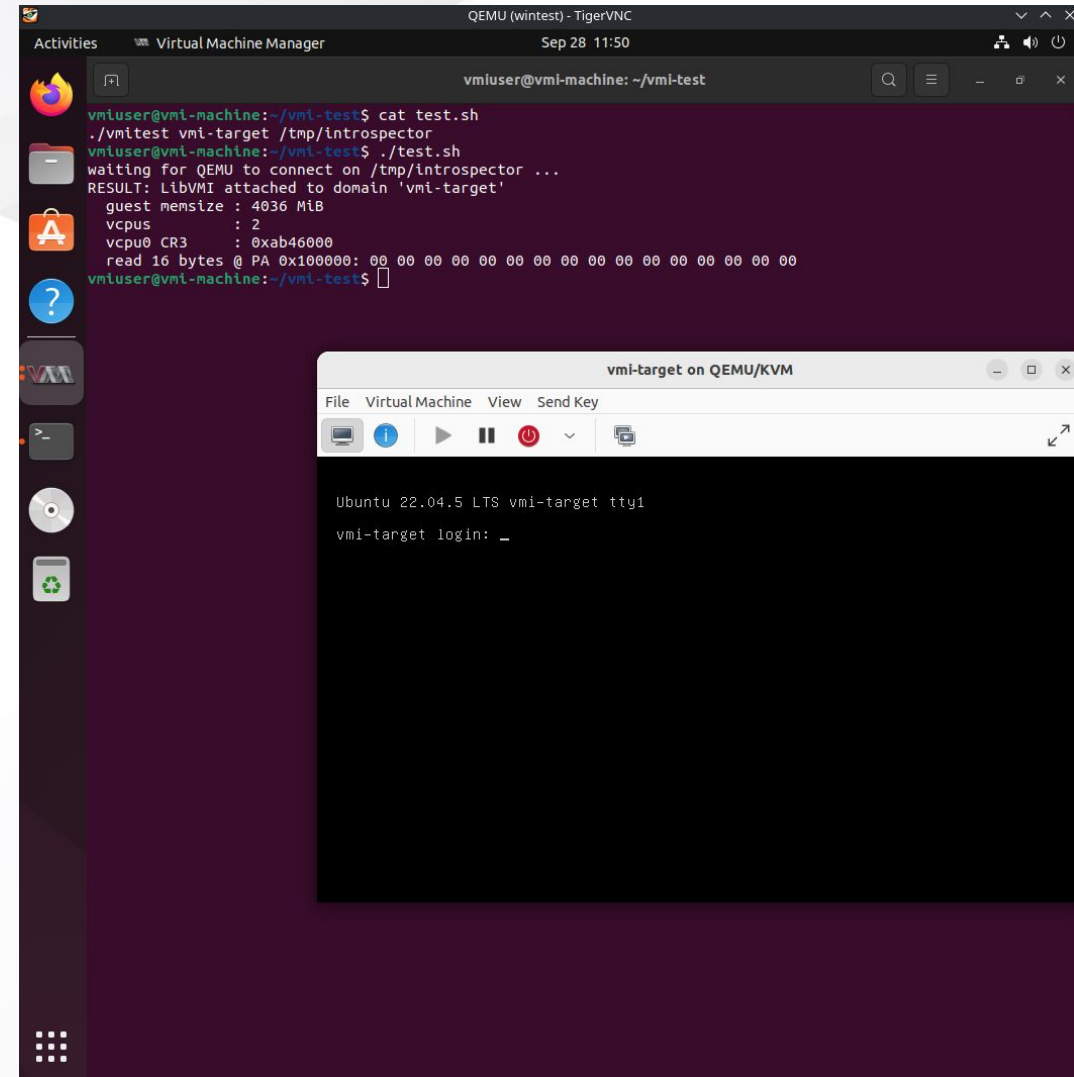
KVMI VM

- To get introspection working, you need specific software
- Issues:
 - KVM-VMI requires you to compile a modified kernel
 - Said kernel is also old (5.4 - 5.15)
 - Requires modified QEMU
 - Need libvmi and libkvmi libraries
- We can make this easier by providing it in a package



KVMI VM

- Ubuntu 22.04 qcow2 image created and ready for use
- Virtual machine allows the user to introspect on nested VMs
- What is inside:
 - KVM-VMI Linux Kernel 5.4
 - Modified QEMU 4.2.1 installed systemwide
 - Docker
 - Upstream libvmi
 - Bitdefender libkvmi
 - Libvirt with apparmor rules



The screenshot displays a terminal window titled 'QEMU (wintest) - TigerVNC' with a timestamp of 'Sep 28 11:50'. The terminal shows a user 'vmiuser@vmi-machine' in the directory '~/vmi-test' running a script 'test.sh'. The script's output indicates that 'LibVMI' successfully attached to a domain named 'vmi-target'. It lists system details: 'guest memsize : 4036 MiB', 'vcpus : 2', and 'vcpu0 CR3 : 0xab46000'. A memory dump is also shown. In the foreground, a smaller window titled 'vmi-target on QEMU/KVM' shows the 'Ubuntu 22.04.5 LTS vmi-target tty1' login screen with the prompt 'vmi-target login: _'.

KVMI VM

- Result
 - User can download VM image from github releases
 - User is able to run it in QEMU
 - User can introspect on nested virtual machines using RootView or other software utilizing libvmi

Ubuntu 22.04 VMI VM Latest Compare

moradomi released this 2 weeks ago · 28 commits to main since this release · KVMI-VM · 275af69

Credentials for this virtual machine:
User: vmiuser
Password: Ch@ng3M3

Run the following to combine the parts:
`cat ubuntu-vmi.qcow2.zst.part_* > ubuntu-vmi.qcow2.zst`

Copy `/usr/share/ovmf/x64/OVMF_VARS.4m.fd` to the vm directory
Then you can unzstd as normal

▼ Assets 10

ubuntu-vmi.qcow2.zst.part_aa	sha256:1da04fcb86fcd682dee46e...	1	1.86 GB	2 weeks ago
ubuntu-vmi.qcow2.zst.part_ab	sha256:a1d9fb7815893dbdce6c18...	1	1.86 GB	2 weeks ago
ubuntu-vmi.qcow2.zst.part_ac	sha256:87092884b123d753df9c81...	1	1.86 GB	2 weeks ago
ubuntu-vmi.qcow2.zst.part_ad	sha256:55207459a09e9945a5a54e...	2	1.86 GB	2 weeks ago
ubuntu-vmi.qcow2.zst.part_ae	sha256:dc7222fc3a1392fdb2fc7...	2	1.86 GB	2 weeks ago
ubuntu-vmi.qcow2.zst.part_af	sha256:3a8dbc1d29ea3f4ec4389e...	2	1.86 GB	2 weeks ago
ubuntu-vmi.qcow2.zst.part_ag	sha256:b969aff4ad2a16766da7e1...	2	1.12 GB	2 weeks ago
vmi-vm.sh	sha256:1d74f6883ba34a70ad6a79...	3	609 Bytes	2 weeks ago

VM CLI

- Full VM lifecycle CLI
 - create, start, stop, restart, destroy, inspect, list, console
- JSON-based configs

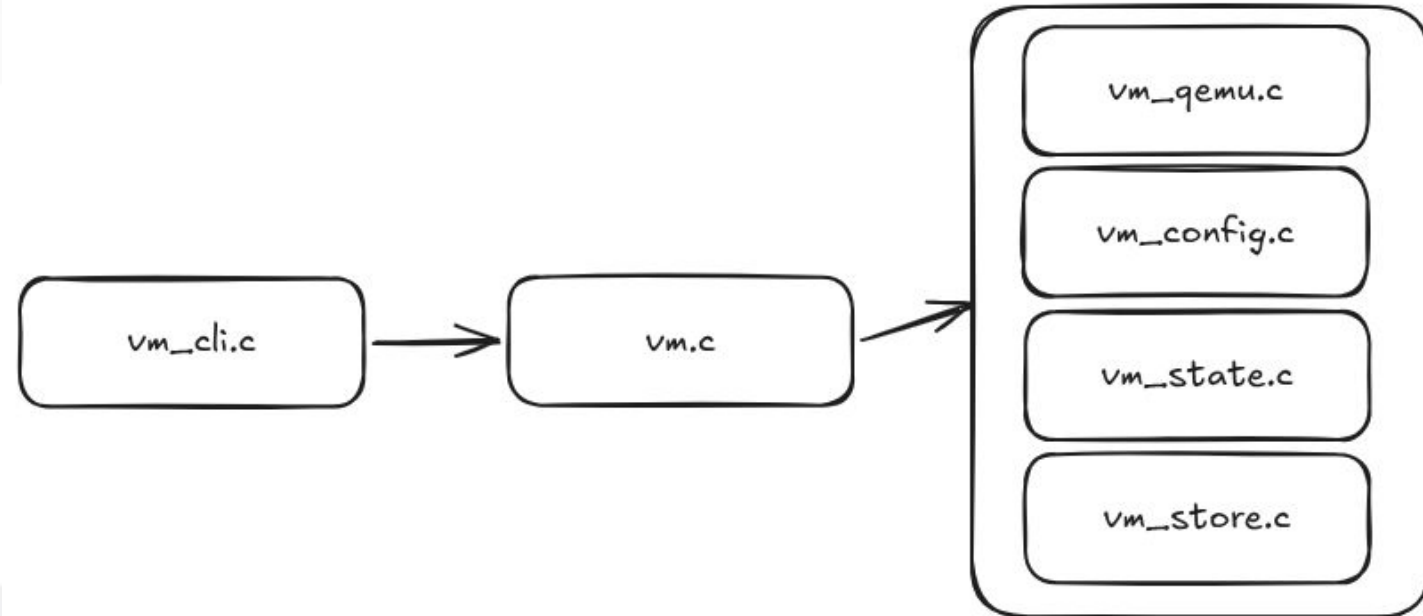
```
vmiuser@vmi-machine:~/Desktop/rootview$ ./rv vm
usage: rv vm <command> [args]

commands:
  create <name> [--config <file>] [--memory <mb>] [--cpus <n>]
               [--disk <path>] [--cdrom <path>] [--network none|user|tap]
               [--display none|gtk|vnc] [--no-kvm] [--no-kvmi] [--extra-args "<args>"]
  start <name>      boot a created vm
  stop <name>       gracefully shut a vm down
  restart <name>    stop then start
  destroy <name> [--force] delete a vm and its state
  inspect <name>    print full vm state
  list             list all known vms
  console <name>   attach to the qemu monitor

vmiuser@vmi-machine:~/Desktop/rootview$ ./rv vm list
NAME      STATUS    PID    MEM    CPUS
test      running   3638    1024    1
vmiuser@vmi-machine:~/Desktop/rootview$ ./rv vm inspect test
name:      test
status:    running
pid:       3638
uptime:    991s
qmp socket: .rootview/vms/test/qmp.sock
monitor:   .rootview/vms/test/monitor.sock
kvmi socket: .rootview/vms/test/kvmi.sock
memory:    1024 MB
cpus:      1
kvm:       yes
disk:      (none)
cdrom:     /home/vmiuser/vms/ubuntu-22.04.5-live-server-amd64.iso
network:   user
display:   gtk
vmiuser@vmi-machine:~/Desktop/rootview$ ./rv vm stop test
stopped 'test'
vmiuser@vmi-machine:~/Desktop/rootview$ ./rv vm list
NAME      STATUS    PID    MEM    CPUS
test      stopped   -      1024    1
vmiuser@vmi-machine:~/Desktop/rootview$
```

VM Internals

- Custom-built JSON parser/serializer
- Clean separation of concerns
 - CLI, lifecycle logic, QEMU process management, state persistence, and on-disk storage each in their own module



VM Safety and Validation

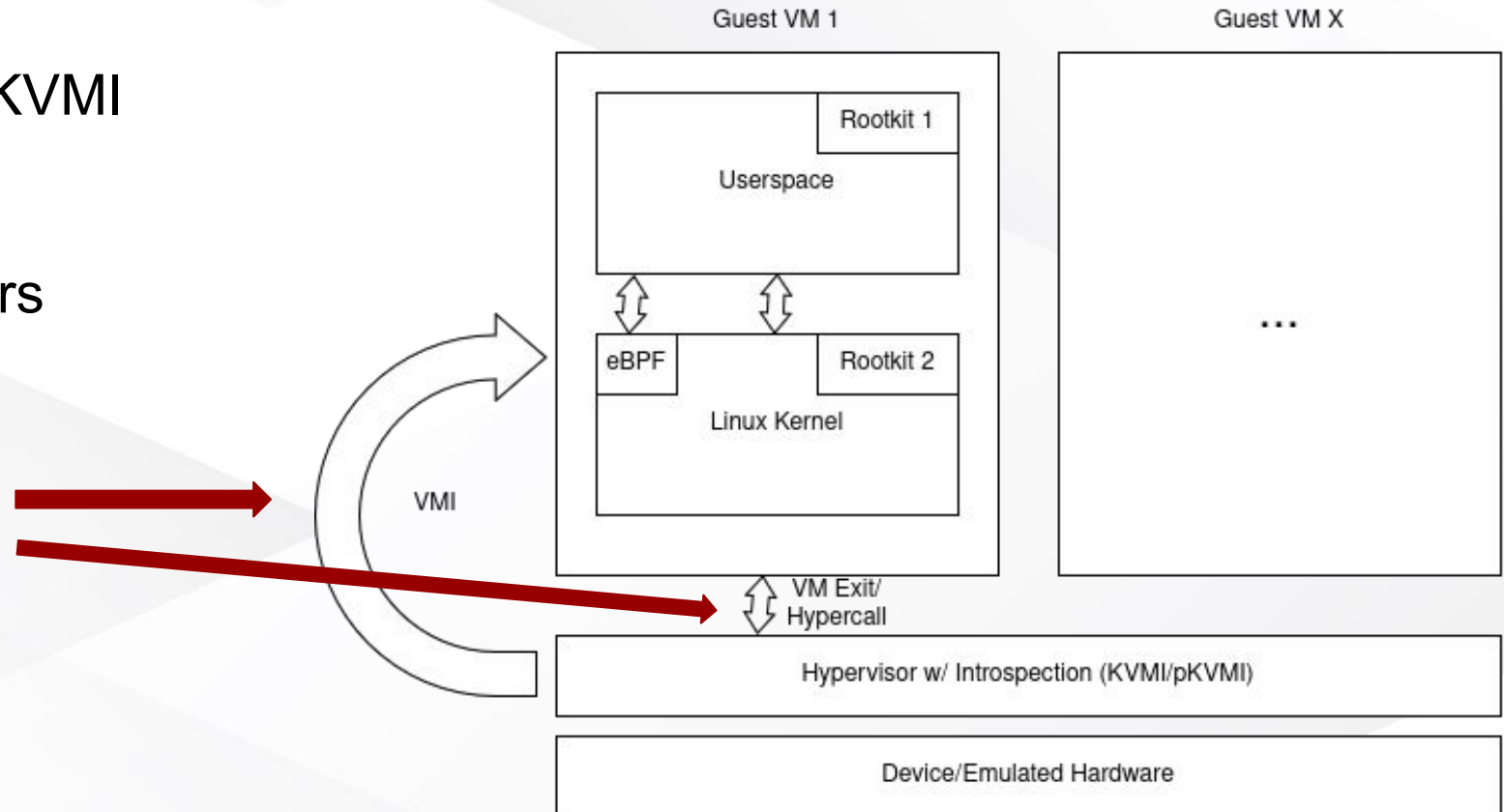
- State-aware: blocks invalid transitions (double-start, destroying a running VM without --force) and detects stale/crashed PIDs
- console command attaches directly to the QEMU monitor over a Unix socket for live debugging
- Validated with an automated smoke test covering every lifecycle transition

```
-- stop --  
[PASS] vm_stop succeeds  
[PASS] vm is stopped after stop  
[PASS] vm_stop rejects a vm that isn't running  
  
-- restart --  
[PASS] vm_restart succeeds from stopped  
[PASS] vm is alive after restart  
  
-- destroy --  
[PASS] vm_destroy without --force rejects a running vm  
[PASS] vm_destroy --force stops and removes the vm  
[PASS] vm_destroy on a missing vm fails  
  
== 14/14 checks passed ==
```

VMI Tooling

- Low-level interfaces with KVM
- R/W GPA and GVA
- R/W guest system registers
- VM-Exit hooks

user can use these



VMI Tooling

system regs

virt mem read

phys mem read

last few entries of the L4
page table
(kernel address space)

```
vmiuser@vmi-machine:~/Desktop/rootview$ ./rv vmi attach test
attached to 'test'. type 'help' for commands, 'quit' to detach.
vmi(test)> help
commands:
  pause          pause the vm
  resume         resume the vm
  rp <paddr hex> <len> read physical memory, hex dump
  wp <paddr hex> <hex bytes> write physical memory
  rv <vaddr hex> <len> read virtual memory (walks the current CR3)
  wv <vaddr hex> <hex bytes> write virtual memory (walks the current CR3)
  reg <name>      read a vcpu register (gp, control, debug, MSR, ...)
  setreg <name> <value hex> write a gp register (rax..r15, rip, rflags) -
                                pause the vm first; this kvmi build has no wire
                                command to write CR/DR/MSR registers. the write
                                only lands on 'resume', 'reg' keeps showing the
                                old value until then
  regs           dump the common registers
  help           show this message
  quit | exit    detach and leave the shell

vmi(test)> pause
paused
vmi(test)> regs
rip=0xffffffff85bc9542 rsp=0xfffffae1540e6fe28 rflags=0x0000000000000046
cr0=0x0000000000005003 cr3=0x000000000e268000 cr4=0x0000000000310ef0
dr6=0x0000000000000000 dr7=0x0000000000000000
idtr_base=0xfffffe0000000000 idtr_limit=0x0fff gdt_base=0xfffffe2e2e48a000 gdt_limit=0x007f
msr_efer=0x00000000000000d01 msr_star=0x0023001000000000 msr_lstar=0xffffffff85c00080 msr_cstar=0xffffffff85c019e0
vmi(test)> rv 0xffffffff85bc9542 16
ffffffff85bc9542 b9 01 00 00 00 89 c2 66 90 e9 f0 8e 23 00 65 8a .....f....#.e.
vmi(test)> rp 0x000000000e268000 64
000000000e268000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000000000e268010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000000000e268020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000000000e268030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
vmi(test)> rp 0x000000000e268f80 128
000000000e268f80 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000000000e268f90 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
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000000000e268fd0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000000000e268fe0 67 b0 fa 3f 00 00 00 00 00 00 00 00 00 00 00 00 g..?.....
000000000e268ff0 67 10 3c 34 00 00 00 00 67 50 c1 33 00 00 00 00 g.<4....gP.3....
vmi(test)>
```

Exit Hook Tooling

Hooks on:

KVM_EVENT_CPUID

KVM_EVENT_DESC

KVM_EVENT_PF
(EPT access)

```
vmiuser@vmi-machine:~/Desktop/rootview$ ./rv hook attach test
attached to 'test'. type 'help' for commands, 'quit' to detach.
hook(test)> help
commands:
  watch <name> <cr0|cr3|cr4|msr_all>    hook writes to a register
  bp <name> <vaddr hex>                  plant a breakpoint hook
  cpuidhook <name> <leaf hex | any>      hook cpuid execution
  deshook <name> <idtr|gdtr|ldtr|tr>     hook a descriptor table load/store
  memhook <name> <vaddr hex> <rw>       hook r/w/x access to a page
  unhook <name>                          remove a hook
  list                                   list active hooks
  listen [ms]                            poll for hook events (default 1000ms, 0 drains only)
  pause                                  pause the vm
  resume                                 resume the vm
  help                                   show this message
  quit | exit                             detach and leave the shell
hook(test)> █
```

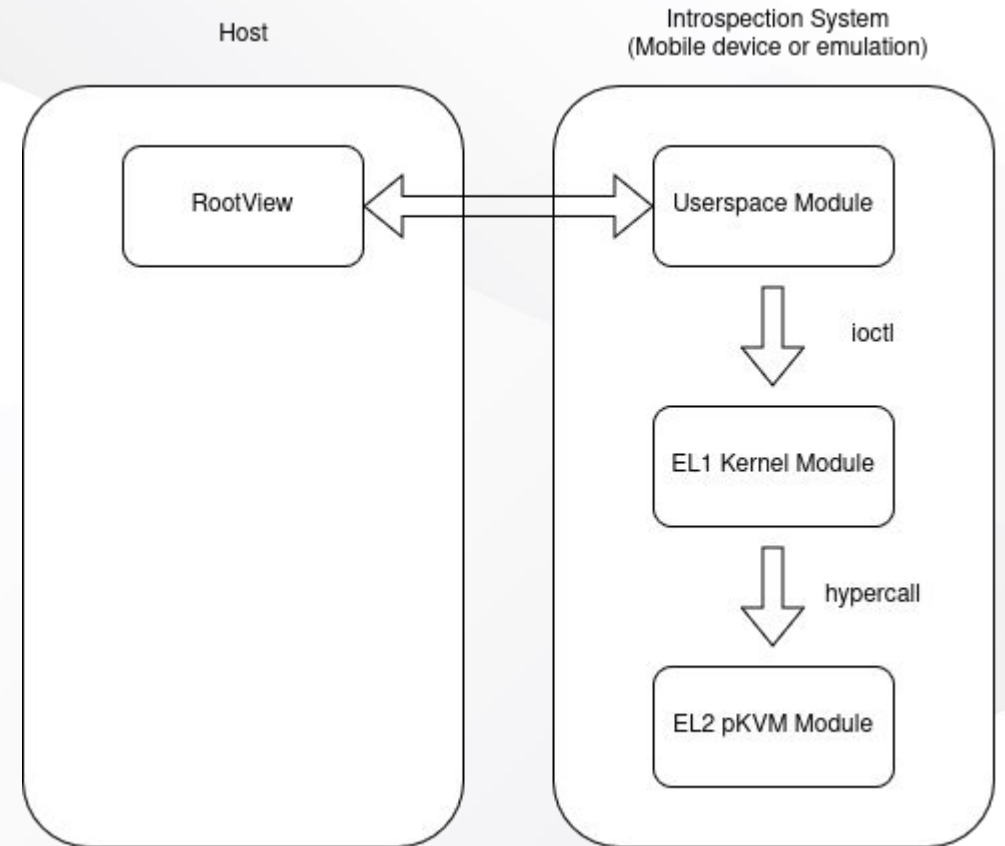
pKVM tooling???

- a lot of newer android phones use pKVM in EL2
- this is a fairly new implementation (2022/2023)
- no tools like KVM exist for it



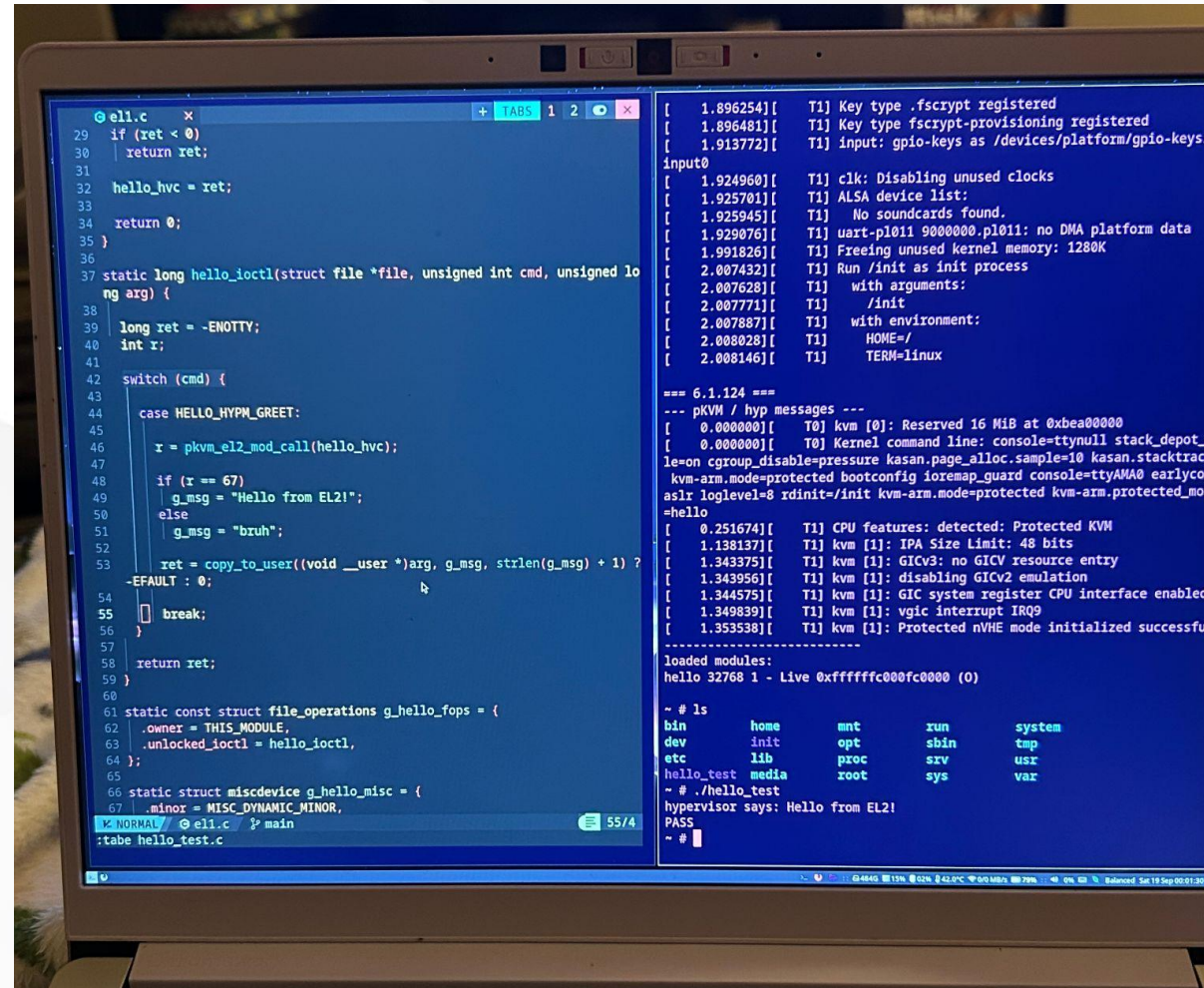
pKVM tooling???

- luckily unlocked pixel phones allow us to write modules (usually its for hardware manufacturers)
- the goal is to write a module that will give us introspection
- circumvents the whole purpose of pKVM
- it is not safe for non-research devices



Hello from EL2!!

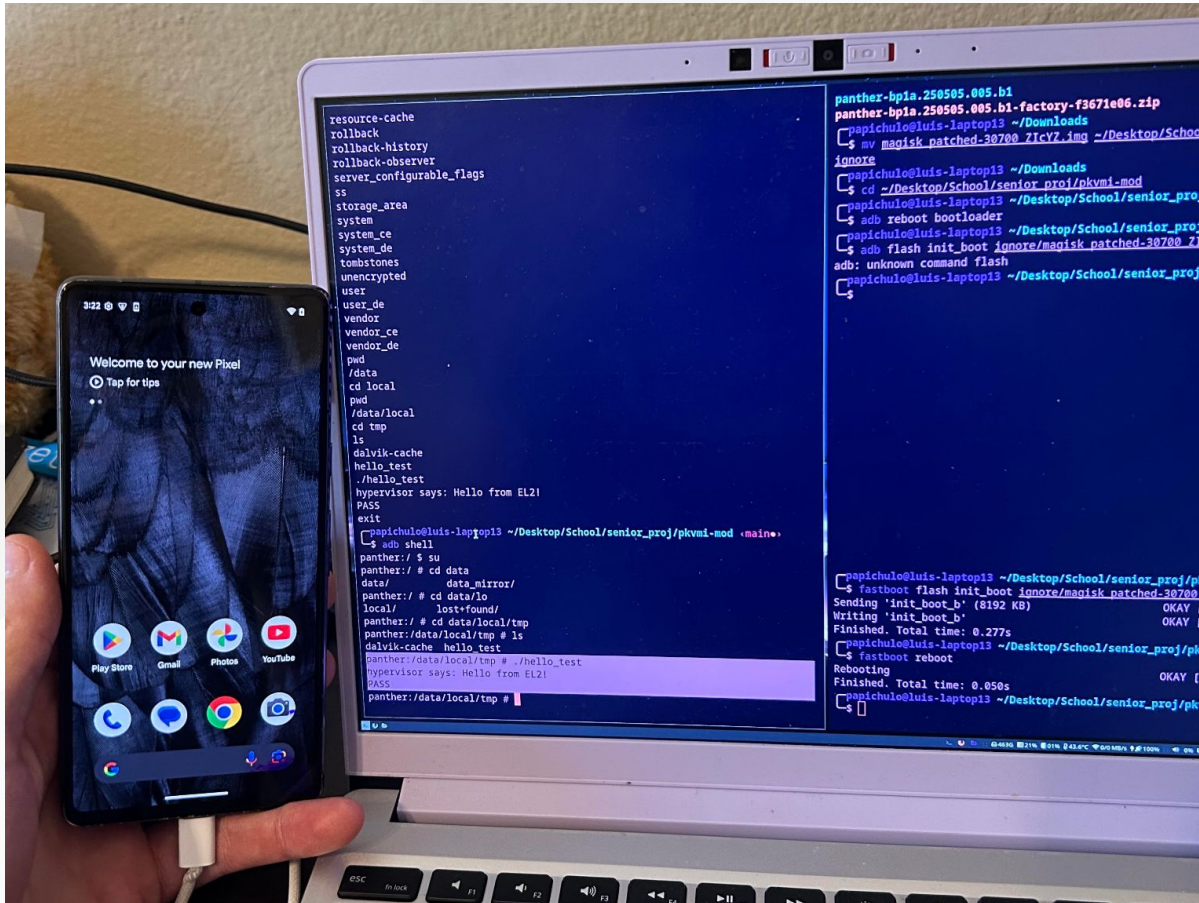
Emulation for development



```
ell.c x + TABS 1 2 x
29 if (ret < 0)
30     return ret;
31
32 hello_hvc = ret;
33
34 return 0;
35 }
36
37 static long hello_ioctl(struct file *file, unsigned int cmd, unsigned long arg) {
38
39     long ret = -ENOTTY;
40     int r;
41
42     switch (cmd) {
43
44     case HELLO_HYPM_GREET:
45
46         r = pkvm_el2_mod_call(hello_hvc);
47
48         if (r == 67)
49             g_msg = "Hello from EL2!";
50         else
51             g_msg = "bruh";
52
53         ret = copy_to_user((void __user *)arg, g_msg, strlen(g_msg) + 1) ?
54             -EFAULT : 0;
55         break;
56     }
57     return ret;
58 }
59
60 static const struct file_operations g_hello_fops = {
61     .owner = THIS_MODULE,
62     .unlocked_ioctl = hello_ioctl,
63 };
64
65 static struct miscdevice g_hello_misc = {
66     .minor = MISC_DYNAMIC_MINOR,
67     .name = "hello_test",
68 };
69
70 int main(void) {
71     register_file_operations(&g_hello_fops);
72     misc_register(&g_hello_misc);
73     return 0;
74 }
75
76 #endif
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Hello from EL2!!

Works on a real
Pixel 7 phone



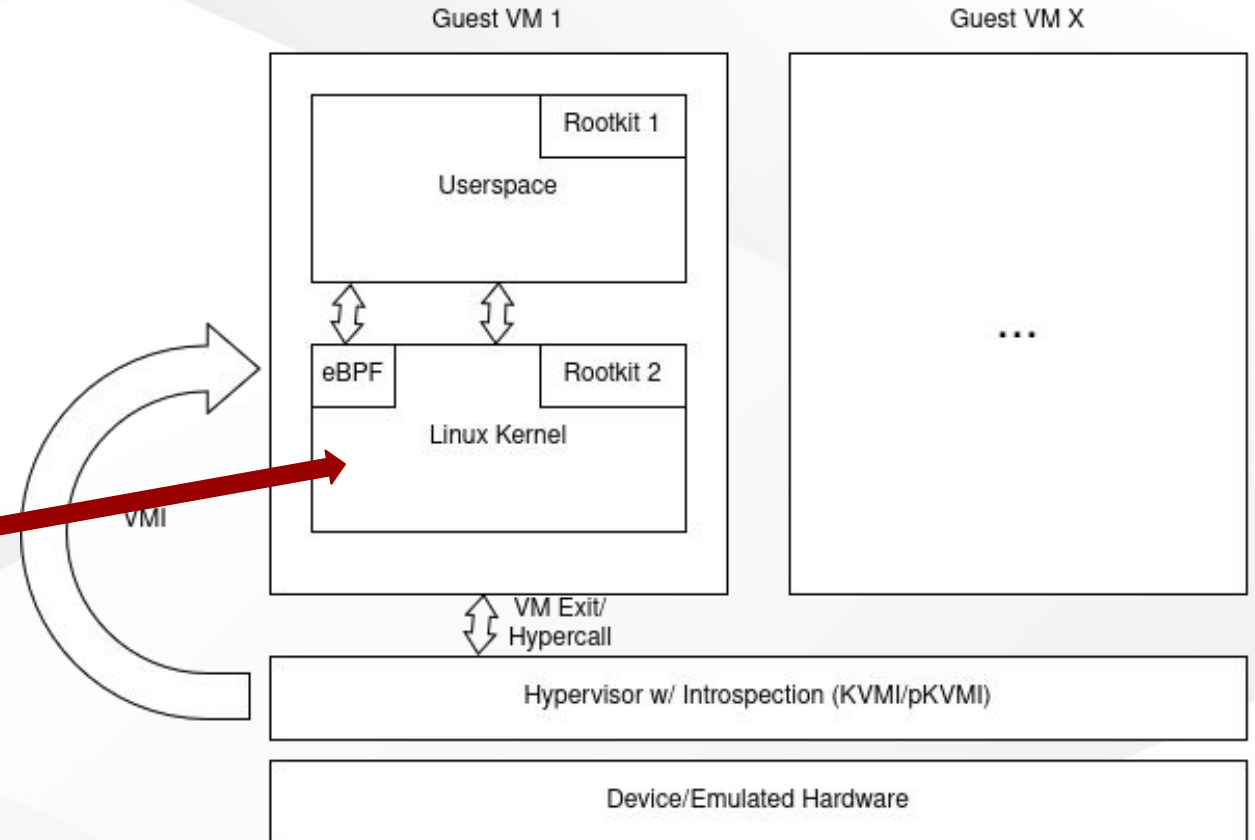
Milestone 1 Progress

Task	Completion	Luis	Braiden	Dominic k	Dylin
Develop VM setup tooling (CLI, JSON configs, introspection integration)	100%	10%	45%	45%	0%
Develop web server for users	100%	0%	0%	0%	100%
Write LibVMI introspection backend	100%	100%	0%	0%	0%
Requirements Document	100%	25%	25%	25%	25%
Design Document	100%	25%	25%	25%	25%
Test Plan	100%	25%	25%	25%	25%

Milestone 2

- Expand capabilities via pKVM tooling
- OS and eBPF Analysis
- Web UI integration with M1 work

“what is going on here?”



Milestone 2 Responsibilities

Task	Luis	Braiden	Dominick	Dylin
pKVM module VMI	100%	0%	0%	0%
Linux Kernel Introspection	10%	90%	0%	0%
Initial eBPF Introspection	10%	0%	90%	0%
Initial Python Bindings	0%	0%	0%	100%
Bindings Integrations w/ web UI	0%	0%	0%	100%

Questions?